HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2013)

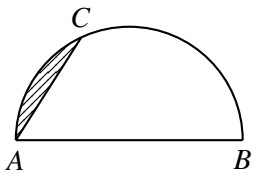
Student No.: 19 Name: _____ Marks: / 22

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. In the figure, the diameter of the semicircle ABC is 3 cm. If chord $AC = 2$ cm, find the area of the shaded region corrected to the nearest $0.01\,\text{cm}^2$.



A. $0.23\,\text{cm}^2$

B. $0.52\,\text{cm}^2$

C. $0.64\,\text{cm}^2$

D. $1.07\,\text{cm}^2$

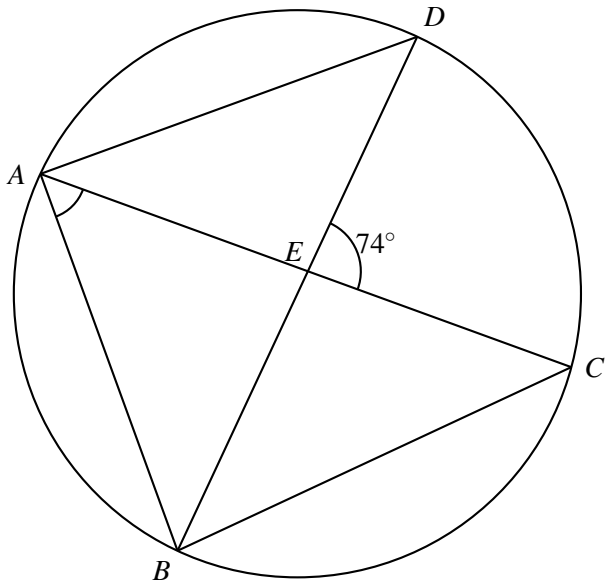
(DSE-2013 PII Q16)

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2. In the figure, $ABCD$ is a circle. AC and BD intersect at E. If $AB = AD$, $AD // BC$, then $\angle BAE =$.



A. 53°

B. 57°

C. 69°

D. 74°

(DSE-2013 PII Q19)

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3. If $0^\circ < x < 90^\circ$, which must be true?

I. $\tan x \tan(90^\circ - x) = 1$

II. $\sin x - \sin(90^\circ - x) < 0$

III. $\cos x + \cos(90^\circ - x) > 0$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2013 PII Q23)

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4. The equation of the circle C is $2x^2+2y^2-4x+8y-5=0$. The coordinates of the points P and Q are $(-1,2)$ and $(4,0)$ respectively. Which of the following is/are true?

I. The radius of C is 5.

II. The midpoint of PQ lies outside C.

III. If G is the centre of C, then $\angle PGQ$ is acute angle.

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2013 PII Q25)

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5. Two numbers are randomly drawn at the same time from seven cards numbered 1, 2, 3, 4, 5, 6 and 7 respectively. Find the probability that the product of the numbers drawn is an odd number.

A. $\frac{2}{7}$

B. $\frac{4}{7}$

C. $\frac{12}{49}$

D. $\frac{16}{49}$

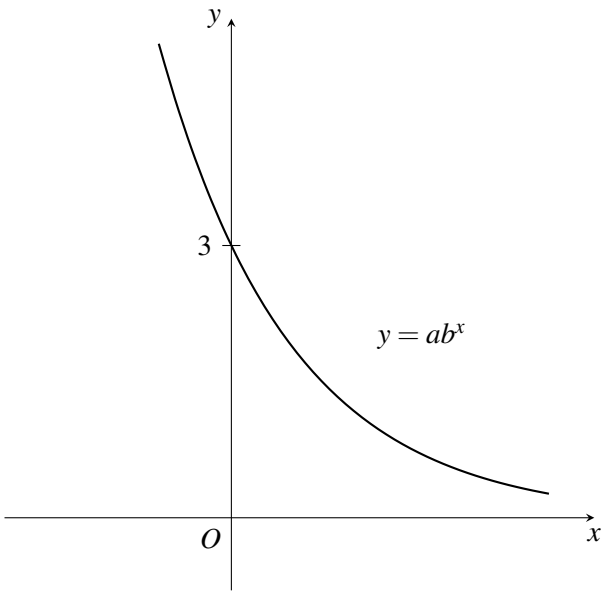
(DSE-2013 PII Q26)

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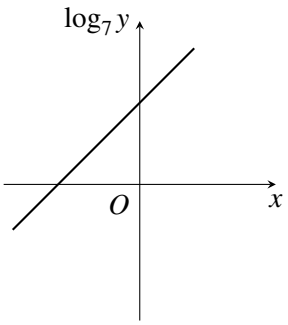
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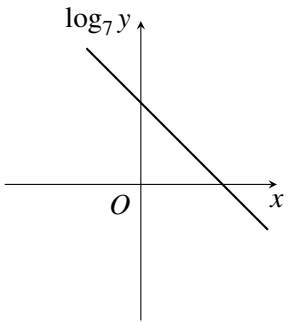
6. The figure below shows the graph of $y = ab^x$, where a and b are constants. Which of the following graphs may represent the relation between x and $\log_7 y$?



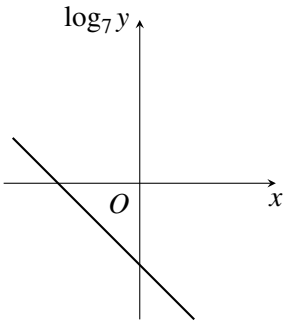
A.



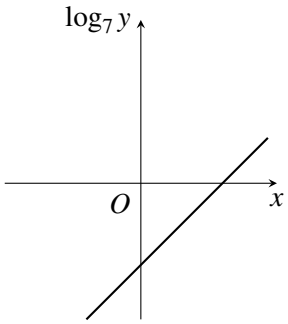
B.



C.



D.



(DSE-2013 PII Q32)

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7. $\text{A00000E00001}_{16} =$

A. $10 \times 16^{11} + 14 \times 16^5 + 17$

B. $11 \times 16^{11} + 15 \times 16^5 + 17$

C. $10 \times 16^{12} + 14 \times 16^6 + 272$

D. $11 \times 16^{12} + 15 \times 16^6 + 272$

(DSE-2013 PII Q33)

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8. If $x - \log y = x^2 - \log y^2 - 10 = 2$, then $y =$

A. 100

B. 2 or -4

C. $\frac{1}{100}$ or 10000

D. $\frac{1}{10000}$ or 100

(DSE-2013 PII Q34)

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9. The nth term of a sequence is $2n - 19$. Which are true?

I. 25 is a term of the sequence..

II. The sequence has 10 negative terms.

III. The sum of the first n terms of the sequence is $n^2 - 18n$.

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2013 PII Q38)

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10. If the height of a regular tetrahedron has height 2 cm. then the volume of the tetrahedron is

A. $2\,\text{cm}^3$

B. $\sqrt{3}\,\text{cm}^3$

C. $\sqrt{6}\,\text{cm}^3$

D. $3\sqrt{3}\,\text{cm}^3$

(DSE-2013 PII Q40)

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11. Let O be the origin. If the coordinates of the points A and B are $(0, 12)$ and$(30, 12)$ respectively, then the $y-$coordinate of the circumcentre of ΔOAB is

A. 6.

B. 8.

C. 12.

D. 15.

(DSE-2013 PII Q43)

I can finish _____ questions

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